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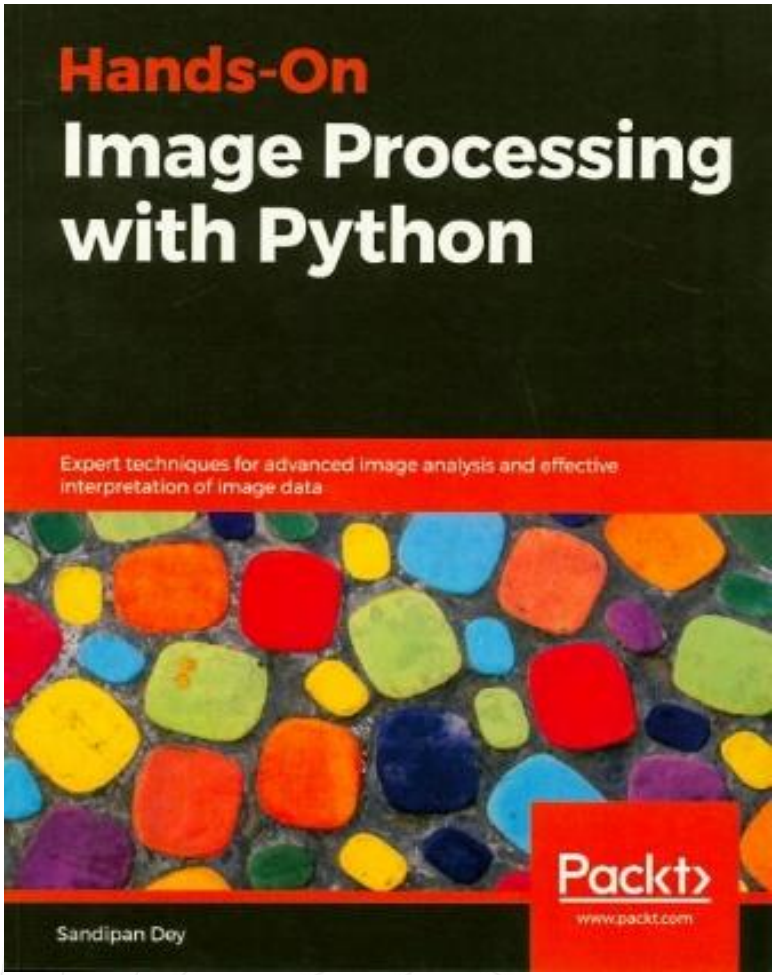
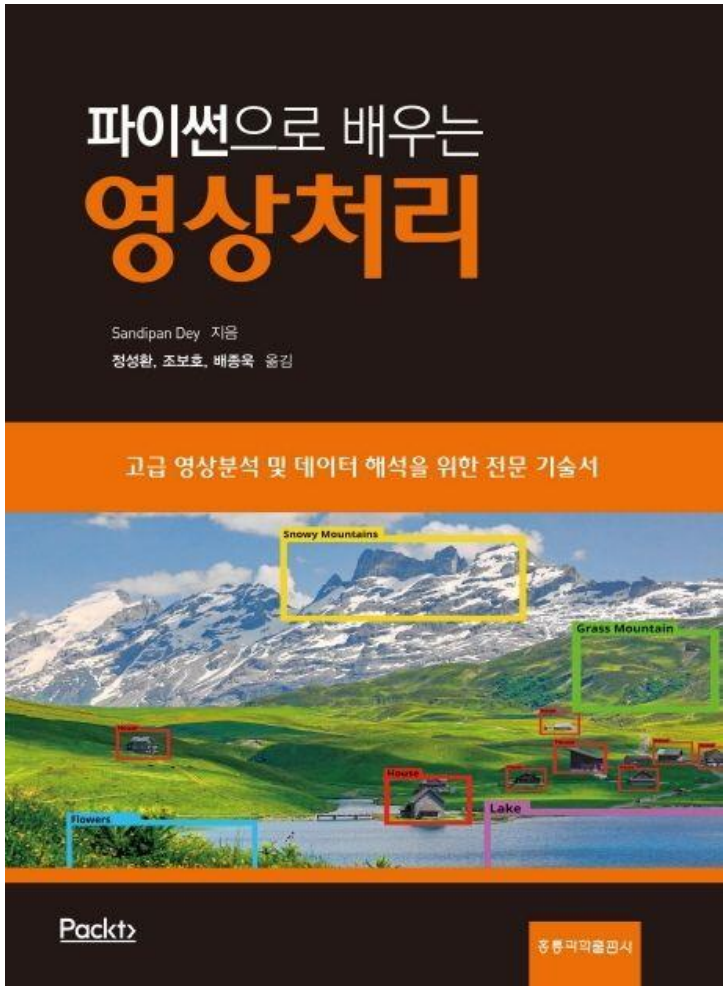
응용수치해석및 실습 제1주

권기운

교재


Hands-On
Image Processing with Python
Expert techniques for advanced image analysis and effective interpretation of image data
Packt
www.packt.com
Sandipan Dey


파이썬으로 배우는
영상처리
Sandipan Dey 지음
정성환, 조보호, 배종욱 옮김
고급 영상분석 및 데이터 해석을 위한 전문 기술서
Snowy Mountains
Grass Mountain
House
Lake
Flowers
Packt
홍릉과학출판사

주차별 강의

주차별수업내용

No	F	강의주차	강의내용	강의활동유형
1		1주차	Chapter 1. Getting started with Image Processing I 9/3: Pre-class (Movie Watching) 9/5 In-class (Science Building AH room)	강의+실습
2		2주차	Chapter 1. Getting started with Image Processing II 9/10: Pre-class (Movie Watching) 9/12 In-class (Science Building AH room)	강의+실습
3		3주차	Chapter 2. Sampling, Fourier Trasformation and Convolution 9/17: Pre-class (Movie Watching) 9/19 In-class (Science Building AH room)	강의+실습
4		4주차	Chapter 3. Convolution and Frequency Domain Filtering 9/24: Pre-class (Movie Watching) 9/26 In-class (Science Building AH room)	강의+실습
5		5주차	Chapter 4. Image Enhancement 10/3: Pre-class (Movie Watching) 10/1 In-class (Science Building AH room)	강의+실습
6		6주차	Chapter 5. Image Enhancement using Derivatives 10/8: Pre-class (Movie Watching) 10/10 In-class (Science Building AH room)	강의+실습
7		7주차	Chapter 6. Morphological Image Processing 10/15: Pre-class (Movie Watching) 10/17 In-class (Science Building AH room)	강의+실습
8		8주차	Midterm Examnination 10/22 Midterm Exam. 10/24 Solution & Claim for the exam	시험

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주차별 강의2

	9		9주차	Chapter 7. Extracting Image Features and Descriptors 10/29: Pre-class (Movie Watching) 10/31 In-class (Science Building AH room)	강의+실습
	10		10주차	Chapter 8. Image Segmentation 11/5: Pre-class (Movie Watching) 11/7 In-class (Science Building AH room)	강의+실습
	11		11주차	Chapter 9. Classical Machine Learning Methods in Image Processing 11/12: Pre-class (Movie Watching) 11/14 In-class (Science Building AH room)	강의+실습
	12		12주차	Chapter 10. Deep Learning in Image Processing 11/19: Pre-class (Movie Watching) 11/21 In-class (Science Building AH room)	강의+실습
	13		13주차	Chapter 11. Deep Learning in Image Processing 11/26: Pre-class (Movie Watching) 11/28 In-class (Science Building AH room)	강의+실습
	14		14주차	Chapter 12. Additional Problems in Image Processing 12/3: Pre-class (Movie Watching) 12/5 In-class (Science Building AH room)	강의+실습
	15		15주차	Final Term Examination 12/10: Pre-class (Movie Watching) 12/12 In-class (Science Building AH room)	시험

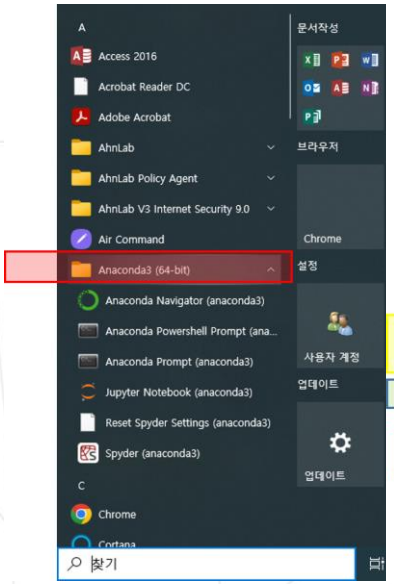
파이썬

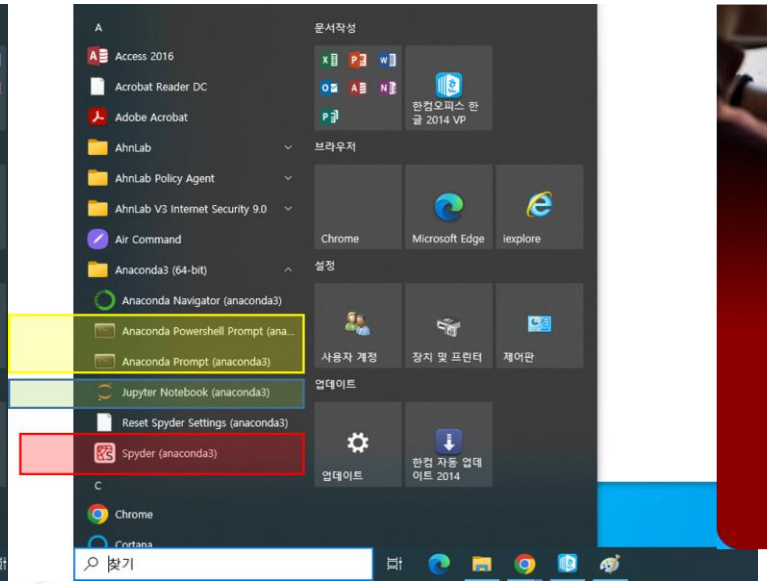
- 프로그래밍 언어
- 최근 심층기계학습(Deep learning)등의 분야에서 활발하게 사용되고 있음
- 윈도우(Windows), 유닉스(Unix), 리눅스(Linux), 맥(Mac) 등의 대부분의 운영체제(OS: Operating System)에서 활용이 가능
- 간결하고 쉬운 문법
- 구글(Google), 인스타그램(Instagram), 넷플릭스(Netflix), 드롭박스(Dropbox)등에서 다른 프로그래밍 언어들과 같이 사용

Anaconda, Spyder

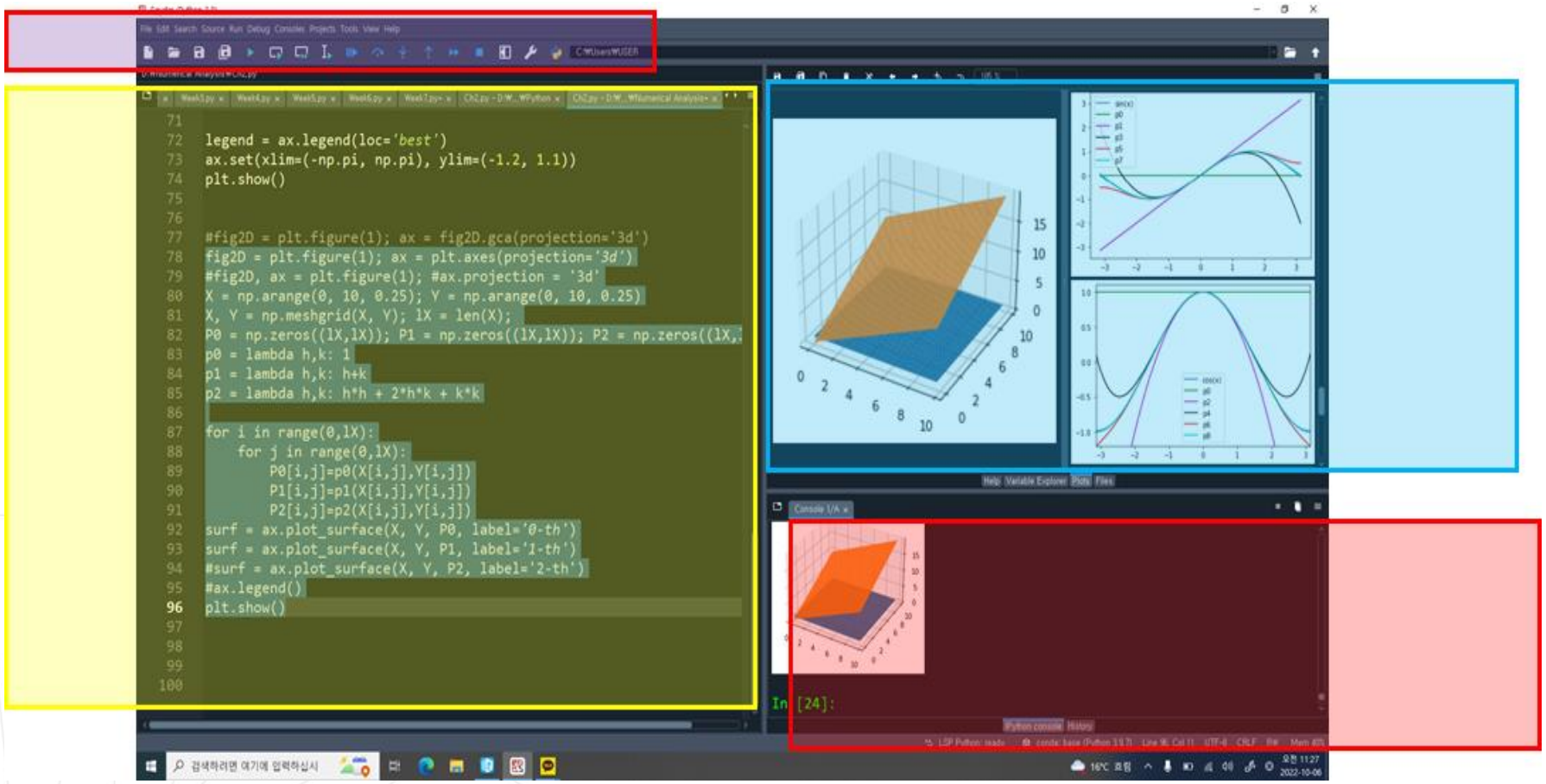
- Continuum Anlaytics에서 개발한 Anaconda 배포판
- 아나콘다는 패키지 관리와 배포를 할 목적으로 만들어진 파이썬이나 R 언어의 공개 소스 배포판
- 패키지 관리 시스템 conda를 통하여 관리된다.
- 파이썬을 설치할 수 있는 홈페이지 주소
  https://www.anaconda.com/download

- 아나콘다가 성공적으로 설치되었다면 다음과 같은 순서로 파이썬 편집기 스파이더(Spyder)에 접근할 수 있다.





Spyder 화면


legend = ax.legend(loc='best')
ax.set(xlim=(-np.pi, np.pi), ylim=(-1.2, 1.1))
plt.show()
#fig2D = plt.figure(1); ax = fig2D.gca(projection='3d')
fig2D = plt.figure(1); ax = plt.axes(projection='3d')
#fig2D, ax = plt.figure(1); #ax.projection = '3d'
X = np.arange(0, 10, 0.25); Y = np.arange(0, 10, 0.25)
X, Y = np.meshgrid(X, Y); lX = len(X);
P0 = np.zeros((lX,lX)); P1 = np.zeros((lX,lX)); P2 = np.zeros((lX,
p0 = lambda h,k: 1
p1 = lambda h,k: h+k
p2 = lambda h,k: h*h + 2*h*k + k*k
for i in range(0,lX):
for j in range(0,lX):
P0[i,j]=p0(X[i,j],Y[i,j])
P1[i,j]=p1(X[i,j],Y[i,j])
P2[i,j]=p2(X[i,j],Y[i,j])
surf = ax.plot_surface(X, Y, P0, label='0-th')
surf = ax.plot_surface(X, Y, P1, label='1-th')
#surf = ax.plot_surface(X, Y, P2, label='2-th')
#ax.legend()
plt.show()
In [24]:

변수 탐색기, 콘솔 탭

Nan	Type	Size	Value
a	int	1	1
b	list	5	[1, 2, 3, 4, 5]
c	tuple	5	(6, 7, 8, 9, 10)
d	Array of int32	(3,)	[11 12 13]
e	set	2	{14, 15}
f	Array of int32	(2, 3)	[[1 2 3] [4 5 6]]

Help Plots Variable Explorer Files

Console 1/A

```
In [14]: f = np.array([1,2,3],[4,5,6])
Traceback (most recent call last):

  File "C:\Users\USER\AppData\Local\Temp/ipykernel_13640/1045093917.py",
line 1, in <module>
    f = np.array([1,2,3],[4,5,6])

TypeError: Field elements must be 2- or 3-tuples, got '4'


In [15]: f = np.array([[1,2,3],[4,5,6]])

In [16]:
```

IPython console History

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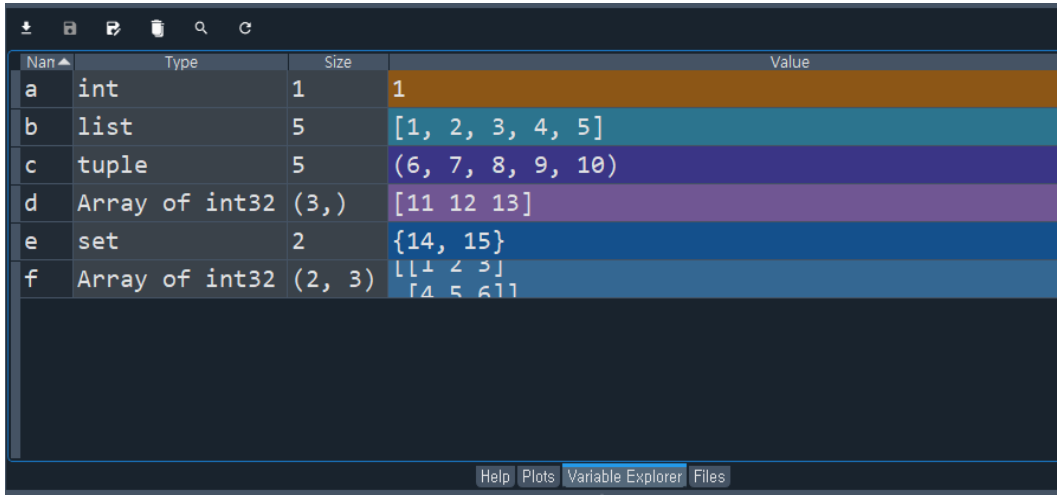
실습 1: 명령 탭, 편집 탭, 변수 탭, 변수, print, 주석

```
# Print
# 명령 탭으로 가서 print("재미있는 영상과 파이썬") 을 입력해 보세요.
print("재미있는 영상과 파이썬")  # 커서를 ) 뒤에 놓고 위의 I 를 눌러 보세요.
print("제 이름은 *** 입니다.")

# 사칙연산
12+25
12-25
12*25
12/25

# 변수 탭에서 생성되는 거 확인해 보세요.
a = 1
g = 1.1
b = [1,2,3,4,5]
c = (1,2,3,4,5)
e = {14,15}
import numpy as np
d = np.array((11,12,13))
f = np.array([[1,2,3],[4,5,6]])

# 리스트, 튜플과 array의 차이를 확인해보세요.
[1,2,3]+[4,5,6]
(1,2,3)+(4,5,6)
np.array((1,2,3))+np.array([4,5,6])

[1,2]+[3,4,5]
np.array([1,2])+np.array([3,4,5])
```



실습 2 : 그림 그리기

```
import matplotlib.pyplot as plt

x = np.linspace(0,2,20)
y = np.cos(np.pi*x)
z = 2*np.sin(np.pi*x)

plt.plot(y); plt.show() #(a)
plt.plot(x,y); plt.show() #(b)
plt.plot(y); plt.plot(x,y) #(c)
plt.plot(x,y,'b-*'); plt.show() #(d)
plt.plot(x,y,'b--*',x,z,'ro'); plt.show() #(e)

plt.plot(x,y,'b--*',x,z,'ro')
plt.xlabel('x (unit:pi)'); plt.title('Cosine and 2*Sine functions')
plt.legend(['cosine','2*sin']); plt.show()  #(f)
plt.plot(y,z); plt.axis('image'); plt.show(); #(g)
plt.plot(y,z); plt.show(); #(h)
```

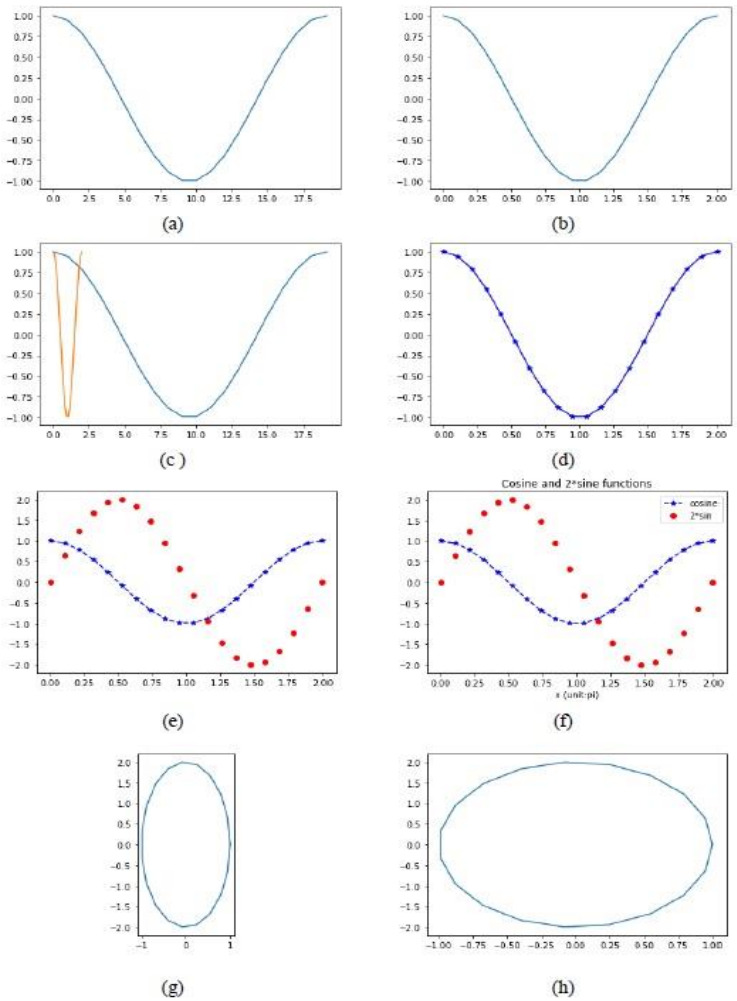
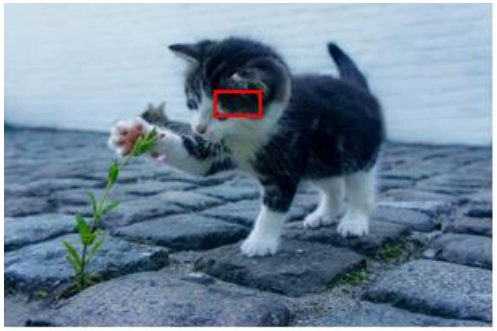
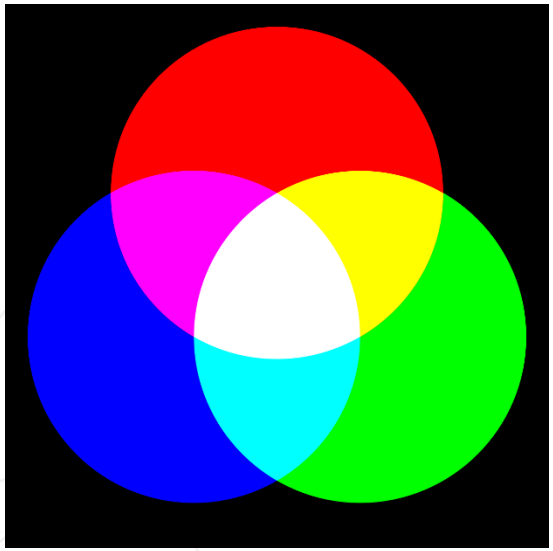


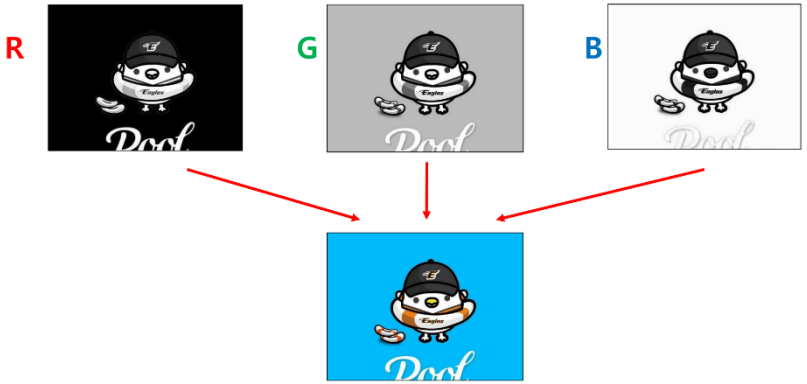
그림 1.7 matplotlib 모듈을 이용한 그림 그리기

영상을 컴퓨터에서 읽으려면



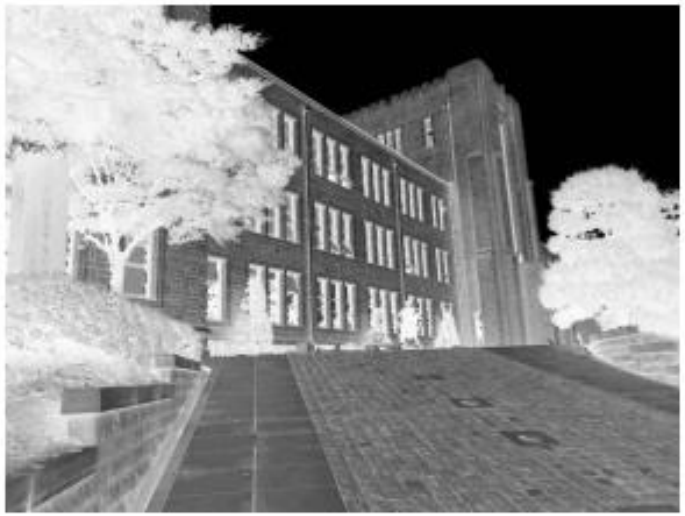
30	10	14	11	10	20	30	50	8
9	1	42	31	70	40	10	15	58
13	89	12	41	120	30	20	30	48
20	11	34	91	20	20	30	20	38
40	110	160	61	90	120	40	70	28
20	210	123	121	11	50	60	50	58
20	140	128	131	30	40	30	150	68
200	210	255	210	190	102	137	189	65





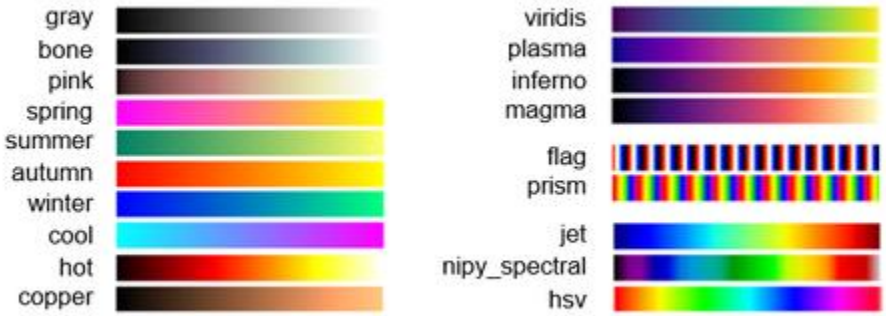
실습 3 : 영상 불러오기 컬러 vs. 흑백

```
import matplotlib.image as mpimg

im = mpimg.imread("MyungJin.jpg")
print(im.shape, im.dtype, type(im))
plt.figure(figsize=(10,10))
plt.imshow(im), plt.axis('off')
plt.show()

imr = im[:,:,0]
plt.gray()
plt.imshow(imr)
plt.axis('off')
plt.show()
```

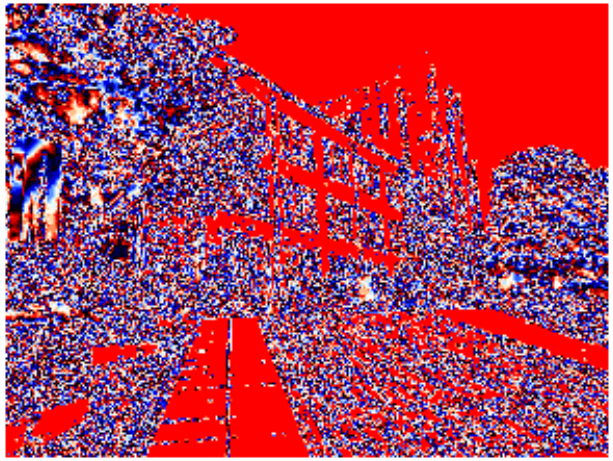


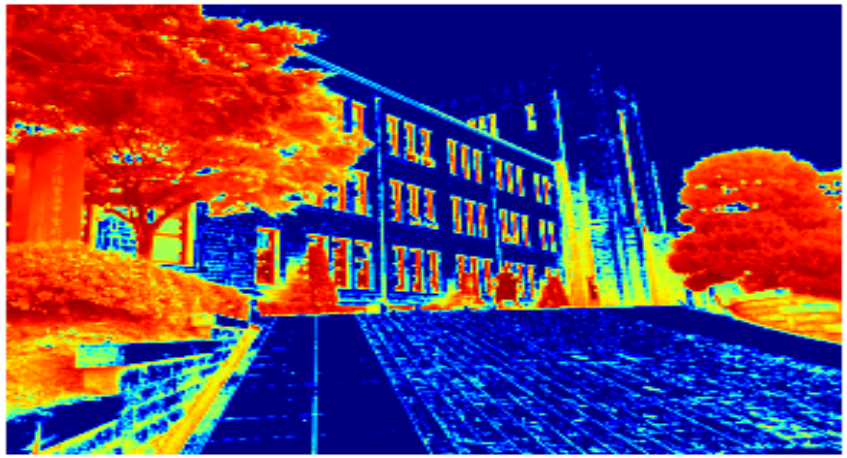
실습 3: Colormap(Color Look-Up Tables:LUT)

- 의사 색상 (Pseudo Color) : 각 픽셀에 흑백 밝기값 (1 차원) 을 할당하는 대신 컬러색상 (3 차원)을 할당한다 .



```
plt.cool(); plt.imshow(imr); plt.axis('off')
plt.flag(); plt.imshow(imr); plt.axis('off')
plt.jet(); plt.imshow(imr); plt.axis('off')
```





컴퓨터가 이해하는 영상

- 영상처리 : 영상을 개선 , 수정 , 변경 또는 분석하는 것과 같이 그림정보를 조절하고 분석하는 것

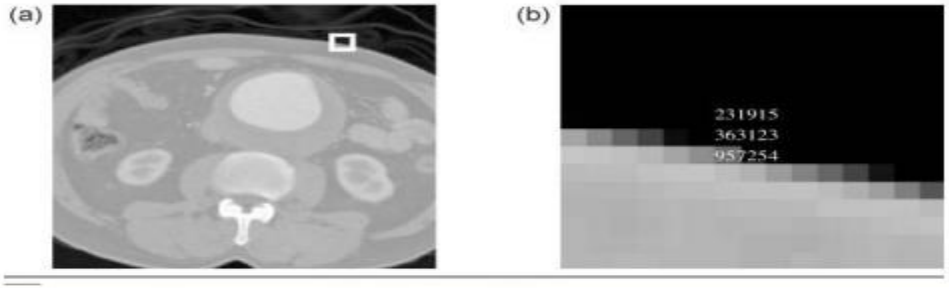
- 아날로그 영상 : 색조나 색상의 연속적 변화 / 안경렌즈를 통한 광학적 처리 (Optical Process)

- 디지털 영상 : 아날로그 영상의 디지털화 (digitization):

1) 표본화 (sampling, 공간적 할당)

2) 양자화 (quantization, 수치적 할당)

- 각 표본 추출 지점 : 픽셀 ➔ 3 차원 경우 voxel
- 의료 영상 : 컴퓨터 단층촬영 (디지털)/ 흉부 X-ray(아날로그)

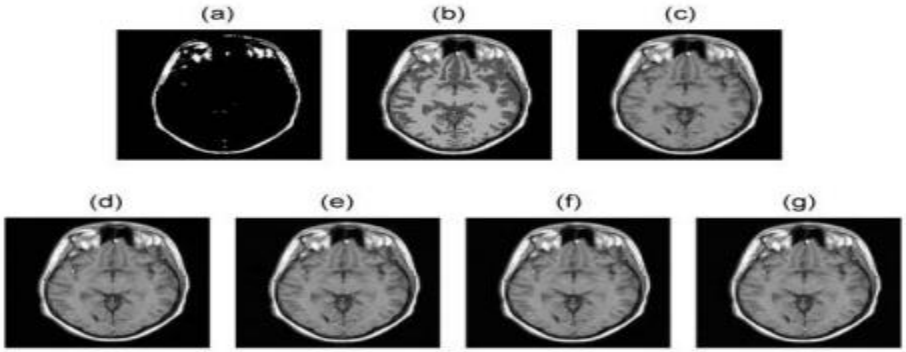


밝기 해상도

- 동적 범위 (Dynamic Range)

  a) 8bit 영상 (0~255) b) 3bit 영상

- 그레이 스케일 영상도가 감소하면 영상이 점차 거칠어진다 .
- 너무 적은 그레이 레벨을 사용할 때 윤곽화(contouring) 현상이 발생한다 .

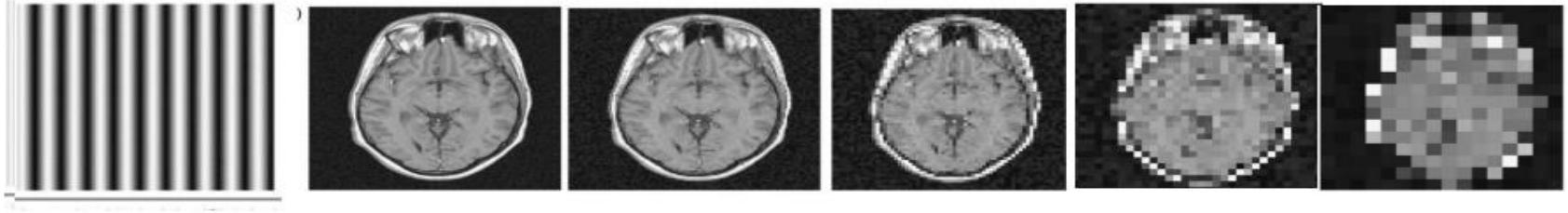


공간해상도

- 디지털화 과정에서 원래 영상의 밝기 값을 나타내기 위해 각 픽셀에 숫자를 할당
- 공간해상도는 취한 표본들이 얼마나 가까이 있는가에 영향을 받는다

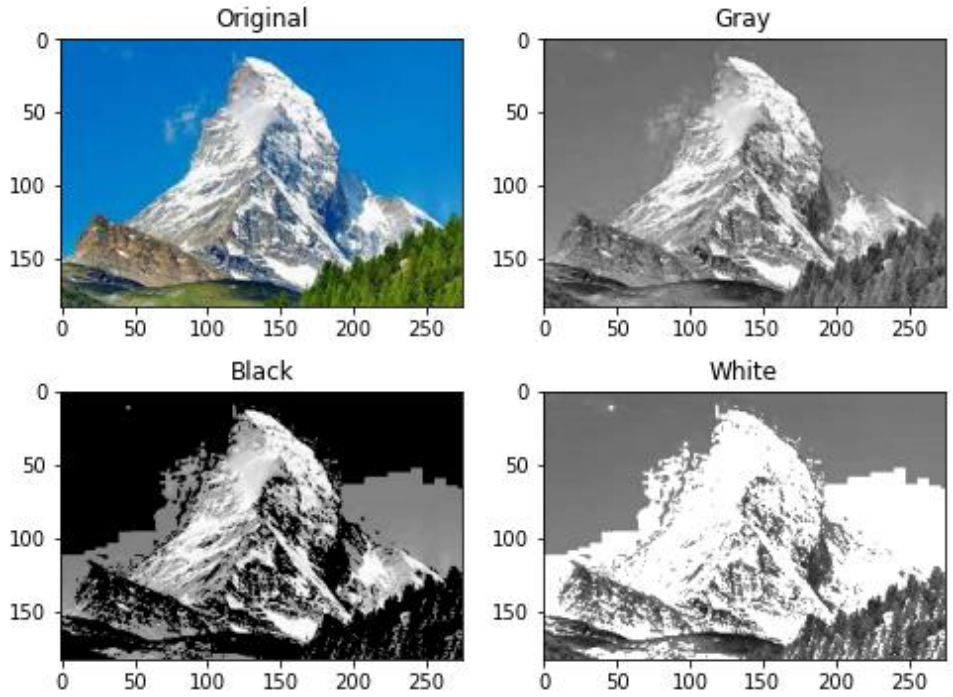
: 공간주파수 (Spatial Frequency)

- 표본화 이론 (Shannon Sampling Theorem) 에 의하면 표현이 가능한 가장 작은 공간 내에서 적어도 2 개 이상의 표본을 얻어야 한다 .
- 디지털 영상에서 분해될 수 있는 가장 높은 공간주파수는 표본화율 (Sampling frequency) 의 절반이 되어야 한다 .
- 픽셀화 (Pixelation)



실습 4: 밝은 영상, 어두운 영상 집중

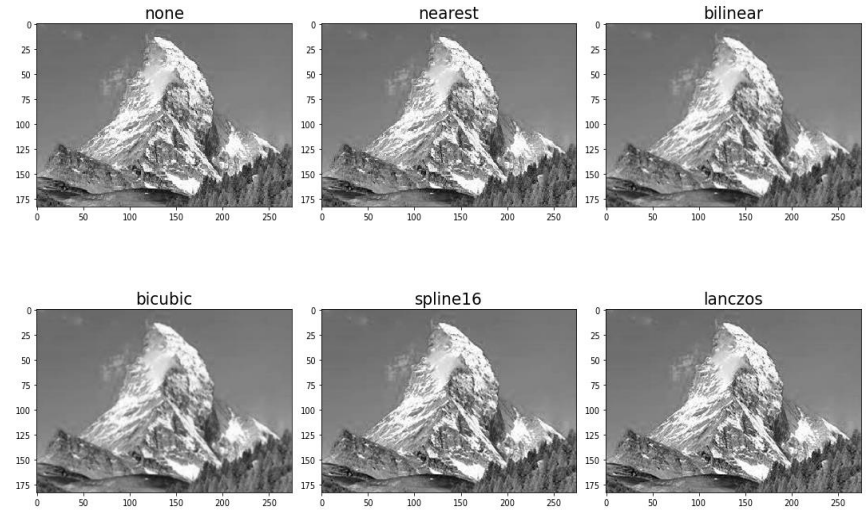
```
im = mpimg.imread("Mountain.png") # im.flags WRITABLE=TRUE 이어야 함
imr = im[:,:,1]; ratio = 0.5

from copy import copy
imblack = copy(imr); imwhite = copy(imr)
if np.max(im) == 1:
    imblack[imr<ratio] = 0
    imwhite[imr>ratio] = 1
elif np.max(im) == 255:
    imblack[imr<ratio*255] = 0
    imwhite[imr>ratio*255] = 255
else:
    print("Error the maximum is", np.max(im))
```



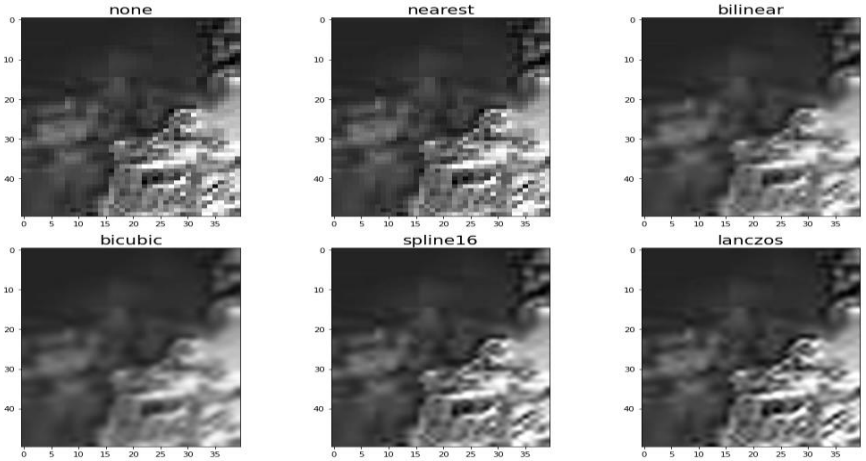

```
plt.figure(figsize=(8,6)); print(im[0,0])
plt.subplot(221), plt.imshow(im), plt.title('Original')
plt.subplot(222), plt.imshow(imr,cmap='gray'), plt.title('Gray')
plt.subplot(223), plt.imshow(imblack,cmap='gray'), plt.title('Black')
plt.subplot(224), plt.imshow(imwhite,cmap='gray'), plt.title('White')
plt.show()
```

실습 5: 보간법

```
im = imr
methods = ['none','nearest','bilinear','bicubic','spline16','lanczos']
fig, axes = plt.subplots(nrows=2,ncols=3, figsize=(15,10))
for ax, interp_method in zip(axes.flat, methods):
    ax.imshow(im,interpolation=interp_method, cmap='gray')
    ax.set_title(str(interp_method), size=20)
plt.tight_layout(); plt.show()
```




```
im = imr[25:75,80:120]
fig, axes = plt.subplots(nrows=2,ncols=3, figsize=(15,10))
for ax, interp_method in zip(axes.flat, methods):
    ax.imshow(im,interpolation=interp_method, cmap='gray')
    ax.set_title(str(interp_method), size=20)
plt.tight_layout(); plt.show()
```



실습 6: RGB, CMY

```
from skimage import color
im = imread("Sohn.png"); im = im[:,:,0:3]; hsv = im
from copy import copy

hsv1 = copy(hsv); hsv2 = copy(hsv); hsv3 = copy(hsv)
plt.figure(figsize=(8,6))
plt.subplot(221), plt.imshow(im), plt.title('Original')
hsv1[:,:,1]=0; hsv1[:,:,2]=0
plt.subplot(222), plt.imshow(hsv1), plt.title('RED')
hsv2[:,:,0]=0; hsv2[:,:,2]=0
plt.subplot(223), plt.imshow(hsv2), plt.title('GREEN')
hsv3[:,:,0]=0; hsv3[:,:,1]=0
plt.subplot(224), plt.imshow(hsv3), plt.title('BLUE')
plt.show()

hsv1 = copy(hsv); hsv2 = copy(hsv); hsv3 = copy(hsv)
plt.figure(figsize=(8,6))
plt.subplot(221), plt.imshow(im), plt.title('Original')
hsv1[:,:,2]=0
plt.subplot(222), plt.imshow(hsv1), plt.title('R+G=YELLOW')
hsv2[:,:,0]=0
plt.subplot(223), plt.imshow(hsv2), plt.title('G+B=CYAN')
hsv3[:,:,1]=0
plt.subplot(224), plt.imshow(hsv3), plt.title('R+B=MAGENTA')
plt.show()
```

실습 7: 영상 조작

```
import numpy as np
from skimage.io import imread
import matplotlib.pylab as plt
from copy import copy
lena = imread("Lenna.png"), lena2 = copy(lena)
lx,ly,_ = lena.shape
X,Y=np.ogrid[0:lx,0:ly]
mask = (X-lx/2)**2 + (Y-ly/2)**2> lx*ly/4
lena[mask,:]=0
plt.figure(figsize=(10,10))
plt.subplot(121), plt.imshow(lena2),
plt.title('Original'), plt.axis('off')
plt.subplot(122), plt.imshow(lena),
plt.title('Mask'), plt.axis('off')
plt.show()
```

실습 8 : alpha 블렌딩 : 메시에서 호나우두

```
import matplotlib.image as mpimg
im1 = mpimg.imread("messi.jpg")/255
im2 = mpimg.imread("ronaldo.jpg")/255
i=1
plt.figure(figsize=(18,15))
for alpha in np.linspace(0,1,20):
    plt.subplot(4,5,i)
    plt.imshow((1-alpha)*im1 + alpha*im2)
    plt.axis('off')
    i += 1
plt.subplots_adjust(wspace=0.05,hspace=0.05)
plt.show()
```

실습 9 : 영상 반전

RGB

RGB:Complement

Red

Red:Complement

그림은 숫자다.

결론 2

컴퓨터는 시키는 일 참 잘한다.
특히, 숫자 계산 참 잘한다.

결론 3

파이썬으로 컴퓨터에게 일을 시킬 수 있다.